

$$\frac{d^3}{dt^3} y(t) + 3 \frac{d^2}{dt^2} y(t) + 2 \frac{d}{dt} y(t) = \frac{d}{dt} u(t) + 2u(t)$$

FDT tra $U(s)$ e $Y(s)$. Condizioni iniziali e derivate nulle

$$(s^3 + 3s^2 + 2s + 0)Y(s) = (s+2)U(s)$$

$$Y(s) = \frac{s+2}{s^3 + 3s^2 + 2s} U(s)$$

calcolare $y(t)$ quando $u(t) = \delta(t)$

$$Y(s) = \frac{s+2}{s^2 + 3s + 2} \quad \text{parziale frast.}$$

$$= \frac{s+2}{s(s+1)(s+2)} = \frac{A_1}{s} + \frac{A_2}{s+1} + \frac{A_3}{s+2}$$

$$A_1 = \lim_{s \rightarrow 0} \frac{s(s+2)}{s(s+1)(s+2)} = 2$$

$$A_2 = \lim_{s \rightarrow -1} \frac{(s+2)(s+2)}{s(s+1)(s+2)} = \frac{-1}{-1} = 1$$

$$A_3 = \lim_{s \rightarrow -2} \frac{(s+2)(s+2)}{s(s+1)(s+2)} = \frac{-2}{-2} = 1$$

$$= \frac{2}{s} + \frac{1}{s+1} + \frac{1}{s+2}$$

$$= 2 - 3e^{-t} + e^{-2t}$$

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -Kx_1^2 - h \sin(x_2) + u \end{cases}$$

$$\odot \quad \theta = \sin(\theta)$$

$K, h > 0$
punti di equilibrio

$$\begin{cases} 0 = x_2 \\ 0 = -Kx_1^2 + u \end{cases} \Rightarrow \begin{cases} \bar{x}_2 = 0 \\ \bar{x}_1^2 = \frac{u}{K} \end{cases} \Rightarrow \begin{cases} \bar{x}_2 = 0 \\ \bar{x}_1 = \pm \sqrt{\frac{u}{K}} \end{cases} \quad u > 0$$

stabilità punti

$$J = \begin{bmatrix} 0 & 1 \\ -2x_1 & -h \cos(x_2) \end{bmatrix}$$

$$J(\sqrt{\frac{u}{K}}, 0) = \begin{bmatrix} 0 & 1 \\ -2\sqrt{\frac{u}{K}} & -h \end{bmatrix} \quad \det(\lambda I - A) = \det \begin{bmatrix} \lambda & -1 \\ 2\sqrt{\frac{u}{K}} & \lambda + h \end{bmatrix}$$

$$= \lambda(\lambda + h) + 2\sqrt{\frac{u}{K}}$$

$$= \underbrace{\lambda^2 + h\lambda + 2\sqrt{\frac{u}{K}}}$$

tutti stesso segno $\Rightarrow \text{Re} < 0 \Rightarrow$ è stabile

$$J(-\sqrt{\frac{u}{K}}, 0) = \begin{bmatrix} 0 & 1 \\ 2\sqrt{\frac{u}{K}} & -h \end{bmatrix} = \det(\lambda I - A) = \det \begin{bmatrix} \lambda & -1 \\ -2\sqrt{\frac{u}{K}} & \lambda + h \end{bmatrix}$$

$$= \lambda(\lambda + h) - 2\sqrt{\frac{u}{K}}$$

$$= \underbrace{\lambda^2 + h\lambda - 2\sqrt{\frac{u}{K}}}$$

cambio di segno $\Rightarrow \text{Re} > 0$, instabile

$$x(k+1) = \begin{bmatrix} 0 & 1 \\ -\frac{1}{2} & -\frac{3}{2} \end{bmatrix} x(k) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(k)$$

$$y(k) = \begin{bmatrix} 1 & 0 \end{bmatrix} x(k) + 0u$$

dire se o sistema é assintoticamente estável, estável, instável

$$\det(\lambda I - A) = \begin{vmatrix} \lambda & -1 \\ \frac{1}{2} & \lambda + \frac{3}{2} \end{vmatrix} = \lambda \left(\lambda + \frac{3}{2} \right) + \frac{1}{2}$$

$$= \lambda^2 + \frac{3}{2}\lambda + \frac{1}{2}$$

$$= \frac{-\frac{3}{2} \pm \sqrt{\left(\frac{3}{2}\right)^2 - 4 \left(\frac{1}{2}\right)}}{2}$$

$$= \frac{-\frac{3}{2} \pm \sqrt{\frac{9}{4} - 2}}{2}$$

$$= \frac{-\frac{3}{2} \pm \sqrt{\frac{5}{4}}}{2}$$

$$= \frac{-\frac{3}{2} \pm \frac{\sqrt{5}}{2}}{2}$$

$$\lambda_{1,2} = \frac{-\frac{3}{2} \pm \frac{\sqrt{5}}{2}}{2} = -\frac{3}{4} \pm \frac{\sqrt{5}}{4}$$

$\lambda_1 = -\frac{3}{4}$ a modulo < 1 é estável

$\lambda_2 = -1$ modulo $= 1$, singular, é estável

$$y(k) = ? \quad u(k) = 0 \quad x(0) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$y(k)$ = evolução livre + evolução forçada

$$x(k) = V \Lambda^k V^{-1} x(0)$$

$$V = \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix}$$

$$= V \begin{bmatrix} \left(-\frac{3}{4}\right)^k & 0 \\ 0 & (-1)^k \end{bmatrix} V^{-1} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\lambda_1 = -\frac{3}{4} \leftarrow (\lambda_1 I - A)u = 0$$

$$= \begin{bmatrix} -\frac{3}{4} & -1 \\ \frac{1}{2} & -\frac{1}{2} + \frac{3}{2} \end{bmatrix} \Rightarrow \begin{bmatrix} -\frac{3}{4} & -1 \\ \frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{cases} -\frac{3}{4}x - y = 0 \\ \frac{1}{2}x + y = 0 \end{cases} \Rightarrow \begin{cases} x + 2y = 0 \\ x + 2y = 0 \end{cases} \Rightarrow \begin{cases} x = -2y \\ x + 2y = 0 \end{cases} \Rightarrow \begin{cases} x = -2y \\ -2y + 2y = 0 \end{cases} \Rightarrow \begin{cases} x = -2y \\ 0 = 0 \end{cases} \Rightarrow \begin{pmatrix} -2 \\ 1 \end{pmatrix} \Rightarrow \begin{pmatrix} -2 \\ 1 \end{pmatrix}$$

$$\lambda_2 = -1 \leftarrow (\lambda_2 I - A)$$

$$= \begin{bmatrix} -1 & -1 \\ \frac{1}{2} & -1 + \frac{3}{2} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{cases} -x - y = 0 \\ \frac{1}{2}x - y = 0 \end{cases} \Rightarrow \begin{cases} -x - y = 0 \\ x + y = 0 \end{cases} \Rightarrow \begin{cases} x = -y \\ -y + y = 0 \end{cases} \Rightarrow \begin{cases} x = -y \\ 0 = 0 \end{cases} \Rightarrow \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$V = \begin{bmatrix} -2 & -1 \\ 1 & 1 \end{bmatrix}$$

$$V^{-1} = \begin{bmatrix} -2 & -1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & -1 & -1 \\ 0 & 1 & 1 & 2 \end{bmatrix}$$

$$y(k) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} -2 \\ 1 \end{bmatrix} \left(-\frac{3}{4}\right)^k + 2 \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} (-1)^k$$

$$= \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} -2 \\ 1 \end{bmatrix} \left(-\frac{3}{4}\right)^k + 2 \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} (-1)^k$$

$$x(k) = V \Lambda^k V^{-1} x(0) = A^k x(0)$$

$$= \begin{bmatrix} -2 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \left(-\frac{3}{4}\right)^k & 0 \\ 0 & (-1)^k \end{bmatrix} \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

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$$= \begin{bmatrix} -2 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \left(-\frac{3}{4}\right)^k & 0 \\ 0 & (-1)^k \end{bmatrix} \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \left(\frac{3}{4}\right)^k & -2(-1)^k \\ -\left(\frac{3}{4}\right)^k & 2(-1)^k \end{bmatrix} =$$

?

$$V = \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}, \quad \tilde{V} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \Rightarrow \tilde{X}(k) = \begin{bmatrix} 2(-\frac{1}{2})^k & 0 \\ 0 & (-1)^k \end{bmatrix}$$

$$x(k) = V \tilde{X} \tilde{V}^T \cdot x(0)$$

$$= \begin{bmatrix} 2(-\frac{1}{2})^k & 1(-1)^k \\ (-1)(-\frac{1}{2})^k & (-1)(-1)^k \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2(-\frac{1}{2})^k + 1(-1)^k & 2(-\frac{1}{2})^k - 2(-1)^k \\ (-1)(-\frac{1}{2})^k + (-1)(-1)^k & (-1)(-\frac{1}{2})^k + 2(-1)^k \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2(-\frac{1}{2})^k - 2(-1)^k \\ (-1)(-\frac{1}{2})^k + 2(-1)^k \end{bmatrix} = x(k)$$

2×2 2×1 2×1

$$C_{11} = 0 +$$

$$y(k) = C x(k) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 2(-\frac{1}{2})^k - 2(-1)^k \\ (-1)(-\frac{1}{2})^k + 2(-1)^k \end{bmatrix} = 2(-\frac{1}{2})^k - 2(-1)^k$$

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